

Homework 1

1. Consider a finite-horizon economy with a representative consumer that has preferences

$$\sum_{t=0}^T \beta^t u(c_t),$$

where $c_t \geq 0, t = 0, \dots, T$ and $u(\cdot)$ is continuously-differentiable, strictly increasing, strictly concave and satisfies $\lim_{c \rightarrow 0} u'(c) = +\infty$. There is a multiple-stage production technology that includes seeding, harvesting, and processing the harvest to make the final good. If $s_t \geq 0$ seeds are planted in period $t - 1$, $h(s_t)$ is harvested in period t . Harvest cannot be consumed but can be used for seeding (which will produce new harvest in the next period) or sent for processing, i.e.

$$s_{t+1} + m_{t+1} \leq h(s_t), t = 0, \dots, T,$$

where $m_{t+1} \geq 0$ is the amount of produce that will be converted into a consumption good in period $t + 1$ using production technology $f(m_{t+1})$. That is,

$$c_t = f(m_t), t = 0, \dots, T.$$

Both $h(\cdot)$ and $f(\cdot)$ are defined on $[0, +\infty)$, continuously differentiable, strictly increasing, weakly concave, and satisfy $h(0) = f(0) = 0$. The initial endowments of seeds s_0 and produce m_0 are given.

- (i) Set up the Planner's decision problem in this economy.
- (ii) Derive the Kuhn-Tucker conditions. Do not forget to indicate which conditions hold in which time periods.
- (iii) Derive the terminal conditions (i.e., characterize how the properties of the endogenous variables in the last period(s)).
- (iv) Derive the difference equation that characterizes the optimal choice and interpret it intuitively.
- (v) Provide an alternative economic interpretation for this decision problem (not related to seeding-harvesting-processing).

2. Consider a finite horizon optimal growth model with CRRA per period utility function $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and Cobb-Douglas production technology $F(K, L) = AK^\alpha L^{1-\alpha}$.

- (a) (15 pts) Briefly explain the idea of the shooting algorithm discussed in class. Describe the numerical procedure you would use to find k_{t+2} which solves the Euler equation for given k_t and k_{t+1} . How will you adjust the guess on k_1 after having computed the associated with this guess sequence $\{k_{t+1}\}_{t=1}^T$; provide a rigorous argument.
- (b) (35 pts) Write a Matlab code which uses the shooting algorithm to compute the optimal capital and consumption profiles and plots them. Set the following parameter values: $\sigma = 2$, $\beta = 0.9$, $\alpha = 0.36$, $\delta = 0.05$, $A = 2$, $L = 1$, $T = 30$, $k_0 = 10$. Your code should be operational for any $\sigma > 0$ (including $\sigma = 1$).
There are many ways of structuring your code. One possibility is to create a function $k_{all}(T, A, \alpha, \delta, L, \sigma, \beta, k_0, k_1)$ that would compute the capital profile for given k_0 and k_1 , and then utilize this function to find the optimal k_1 and the corresponding consumption and capital profiles.
- (c) (5 pts) Suppose that β increases to $\beta = 0.96$. Plot on the same graph the consumption profiles for the old and the new values of β . How do they differ? What's the intuition behind this observation? Produce a similar figure for the capital profiles and provide an intuitive interpretation.
- (d) (5 pts) Suppose that A increases to $A = 3$. How does this change affect consumption and capital profiles? Illustrate it on a graph and explain intuitively.

Submission: Please email your codes to the TA with clear instructions on how to run them. You are allowed to work in teams of two people (in which case make only one electronic submission per team). In addition to the electronic submission, please print out the codes and figures, and attach them to the answers. If working in groups, it's your own responsibility to enforce that all the group members contribute equally.